CHE 565 -- Homework 5

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Introduction

This homework was completed using Python with the Control Systems Library. The report uses the same block-diagram logic as the Simulink assignment, but the systems are constructed programmatically using transfer functions, state-space conversions, and `ct.interconnect`.

The cascade-control problems compare the response with and without the inner feedback loop. The first controller is treated as an ideal PI controller, while the inner controller is P-only. For the no-cascade case, the inner feedback is disconnected and the inner controller gain is set to 1.

$$G_{c,1}(s) = K_{c,1}\left(1 + \frac{1}{\tau_{I,1} s}\right)$$

$$G_{c,2}(s) = K_{c,2}$$

$$IAE = \int_0^T |r(t) - y(t)|\, dt$$

```
    # P-only, but represented as a transfer function so ct.ss() works.
    return ct.tf([Kc], [1])

Gc2 = p_only_controller(Kc2)
Gc2_blk = ct.ss(Gc2, name='Gc2', inputs='E2', outputs='Yc2')%
```

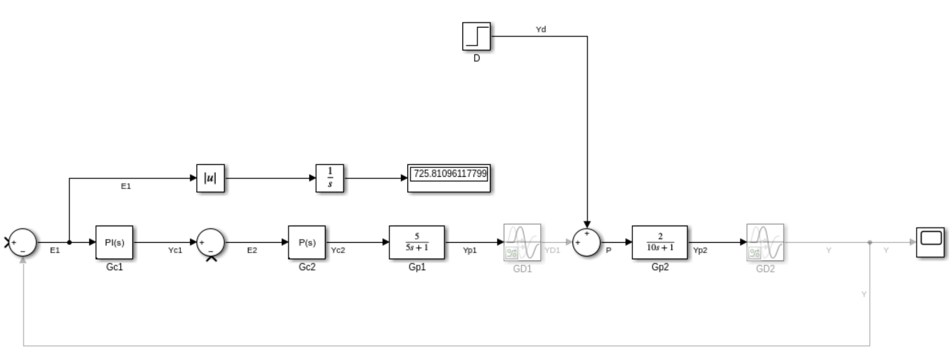


Figure 1: Simulink block diagram for the cascade-control system.

Problems 1--3: Cascade Control

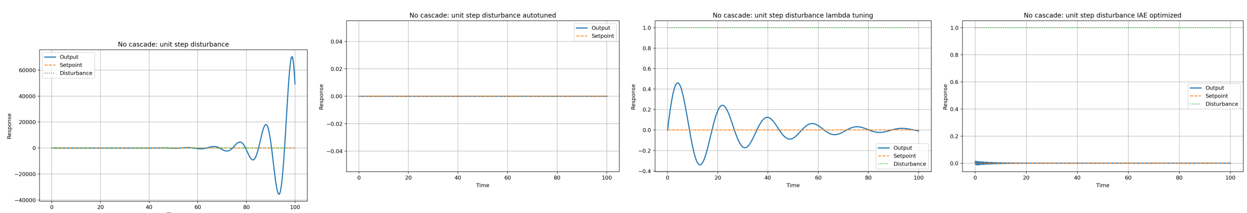
Problem 1: System Without Cascade Control

For the no-cascade case, the inner feedback path was disconnected and the P-only inner controller gain was set to $K_{c,2} = 1$. The outer PI controller was then tuned using the autotuning from simulink and lambda tuning method and minimizing the IAE

$$K_{c,1} = 1.0000, \qquad \tau_{I,1} = 1.0000, \qquad I_1 = 1.0000$$

Case	$K_{c,1}$	$\tau_{I,1}$	IAE
Disturbance	1	1	450862.8604
Autotune	0.1837	12.2474	0.0000
Lambda	0.6094	4.3333	9.0895
IAE optimized	874.3338	1213.2216	0.0793

Table 1: Problem 1 results without cascade control.

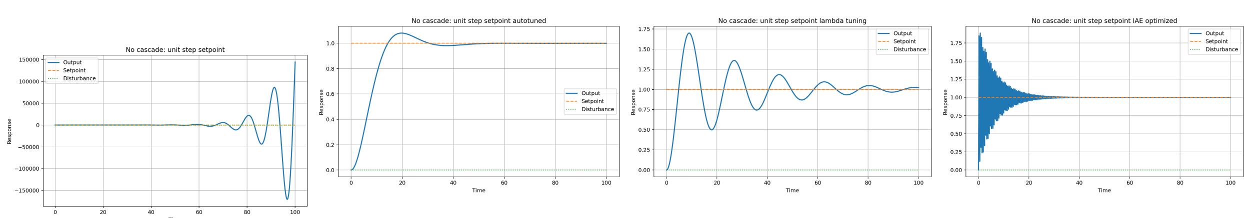


(a) Disturbance response (b) Disturbance response (auto-tuned) (c) Disturbance response (lambda tuning) (d) Disturbance response (IAE optimized)

Figure 2: Problem 1 responses without cascade control.

Case	$K_{c,1}$	$\tau_{I,1}$	IAE
Setpoint	1	1	1233339.1635
Autotune	0.1837	12.2474	8.2136
Lambda	0.6094	4.3333	16.5618
IAE optimized	874.3338	1213.2216	4.2617

Table 2: Problem 1 setpoint results without cascade control.



(a) Setpoint response (b) Setpoint response (auto-tuned) (c) Setpoint response (lambda tuning) (d) Setpoint response (IAE optimized)

Figure 3: Problem 1 responses without cascade control.

Problem 2: System With Cascade Control

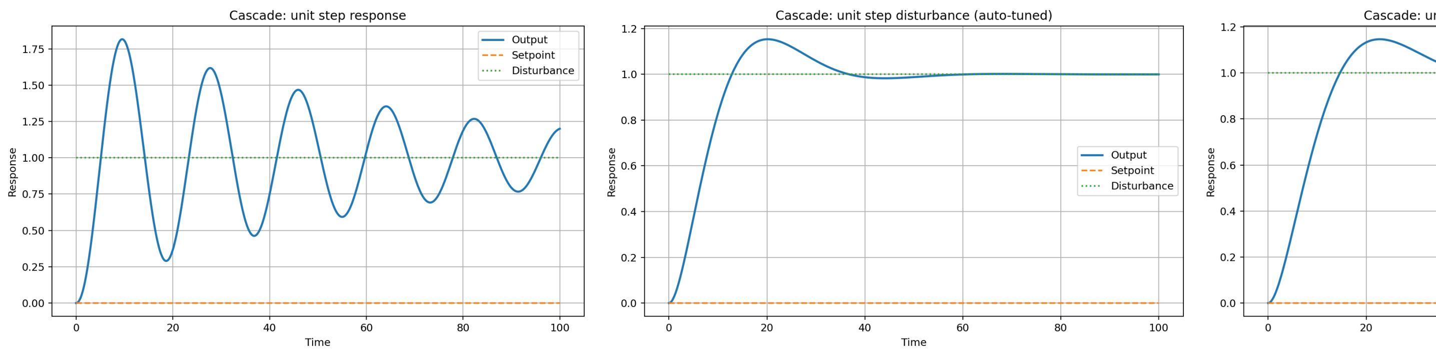
Problem 2: System With Cascade Control

For the cascade case, the inner feedback path was connected and the inner P-only controller gain was set to $K_{c,2} = 0.4$. The outer controller was tuned again because the outer loop sees a different equivalent process.

$$K_{c,1} = 1.0000, \qquad \tau_{I,1} = 1.0000, \qquad I_1 = 1.0000$$

Case	$K_{c,2}$	$K_{c,1}$	$\tau_{I,1}$	IAE
Disturbance	0.4000	1	1.0000	9.8851
Autotune	0.4000	0.7508	4.5461	4.1578
Lambda	0.4000	0.6094	4.3333	4.8484
IAE optimized	0.4000	1378.2789	563.2547	0.0611

Table 3: Problem 2 results with cascade control and unit step disturbance.



(a) Disturbance response

(b) Disturbance response

(c) Dis

Figure 4: Problem 2 responses with cascade control and unit step disturbance.

Case	$K_{c,2}$	$K_{c,1}$	$\tau_{I,1}$	IAE
Setpoint	0.4000	1	1.0000	31.0631
Autotune	0.4000	0.7508	4.5461	8.7692
Lambda	0.4000	0.6094	4.3333	9.7764
IAE optimized	0.4000	1378.2789	563.2547	1.9419

Table 4: Problem 2 results with cascade control and unit step setpoint changes.

Problem 3: Comparison of Cascade and No-Cascade Results

Test	No Cascade IAE	Cascade IAE	Change
Disturbance	450862.8604	9.8851	-450852.9753
Setpoint	0.0000	4.1578	4.1578

Table 5: Comparison of no-cascade and cascade-control performance.

Cascade control is expected to make the largest difference for disturbance rejection. The disturbance enters at the intermediate process variable, so the inner loop can react before the effect fully propagates to the final output. For setpoint tracking, cascade control may still change the response, but the benefit is usually less direct because the setpoint must pass through both the outer and inner loops.

Problem 4: Relative Gain Array for the 4 by 4 Process

$$\Lambda = K \circ \left(K^{-1}\right)^T$$

$$K = \begin{bmatrix} 0.4300 & 0.4300 & 0.2300 & 0.2200 \\ -0.3300 & 0.3200 & -0.2000 & 0.2000 \\ 0.2200 & 0.2300 & 0.4200 & 0.4100 \\ -0.2200 & 0.2200 & -0.3200 & 0.3200 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 0.6795 & 0.7167 & -0.1958 & -0.2004 \\ 0.8768 & 0.8566 & -0.3548 & -0.3786 \\ -0.1885 & -0.2078 & 0.6999 & 0.6964 \\ -0.3678 & -0.3654 & 0.8507 & 0.8826 \end{bmatrix}$$

y_i	u_1	u_2	u_3	u_4
y1	0.6795	0.7167	-0.1958	-0.2004
y2	0.8768	0.8566	-0.3548	-0.3786
y3	-0.1885	-0.2078	0.6999	0.6964
y4	-0.3678	-0.3654	0.8507	0.8826

Table 6: Relative gain array for Problem 4.

Output	Manipulated Variable	λ_{ij}
y1	u2	0.7167
y2	u4	-0.3786
y3	u1	-0.1885
y4	u3	0.8507

Table 7: Suggested pairings for Problem 4.

The suggested pairings are chosen from positive RGA values that are relatively close to one while avoiding negative pairings. Negative relative gains are avoided because they indicate that closing other loops can reverse the apparent gain direction.

Problem 5: Relative Gain Array for the 2 by 2 Process

$$K = \begin{bmatrix} 5 & 2 \\ 3 & 6 \end{bmatrix}$$

$$\Lambda = \begin{bmatrix} 1.2500 & -0.2500 \\ -0.2500 & 1.2500 \end{bmatrix}$$

y_i	u_1	u_2
y1	1.2500	-0.2500
y2	-0.2500	1.2500

Table 8: Relative gain array for Problem 5.

The diagonal pairings are preferred because the diagonal RGA values are positive and greater than one, while the off-diagonal values are negative. Therefore, the best pairing is y1 with u1 and y2 with u2.

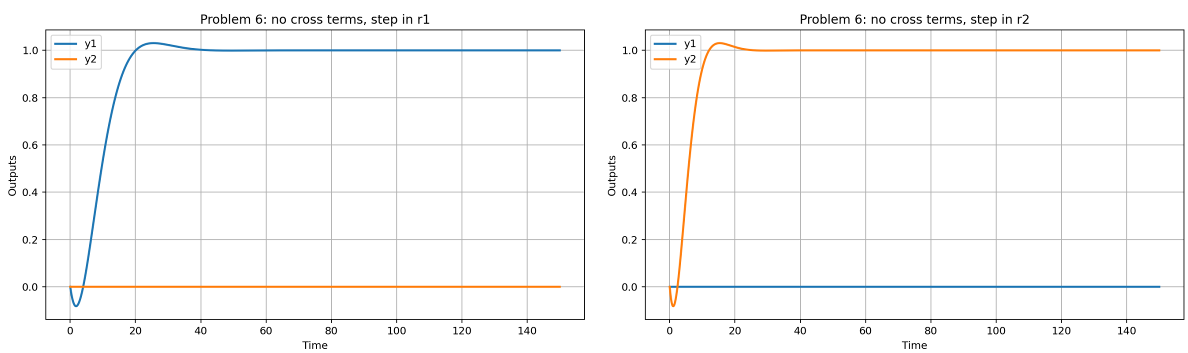
Problems 6--10: MIMO Control, Cross Terms, and Decoupling

Problem 6: Two Single-Loop Controllers with Cross Terms Neglected

$$K_c = \frac{\tau_p}{K_p(\lambda + \theta)}, \qquad \tau_I = \tau_p$$

Loop	K_p	τ_p	θ	λ	K_c	τ_I
G11	5.0000	4.0000	5.0000	5.0000	0.0800	4.0000
G22	6.0000	10.0000	3.0000	3.0000	0.2778	10.0000

Table 9: PI tuning values for the two diagonal loops.



(a) Step in r1

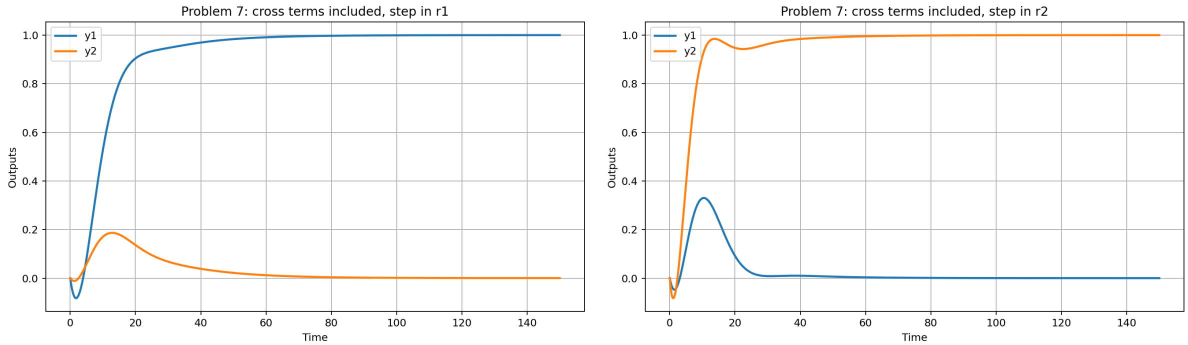
(b) Step in r2

Figure 5: Problem 6 responses with cross terms neglected.

Problem 7: Cross Terms Included

Test	Controlled-output IAE	Interaction area
Step in r1	12.4992	4.5391
Step in r2	7.4995	4.3600

Table 10: Problem 7 performance with cross terms included.



(a) Step in r1

(b) Step in r2

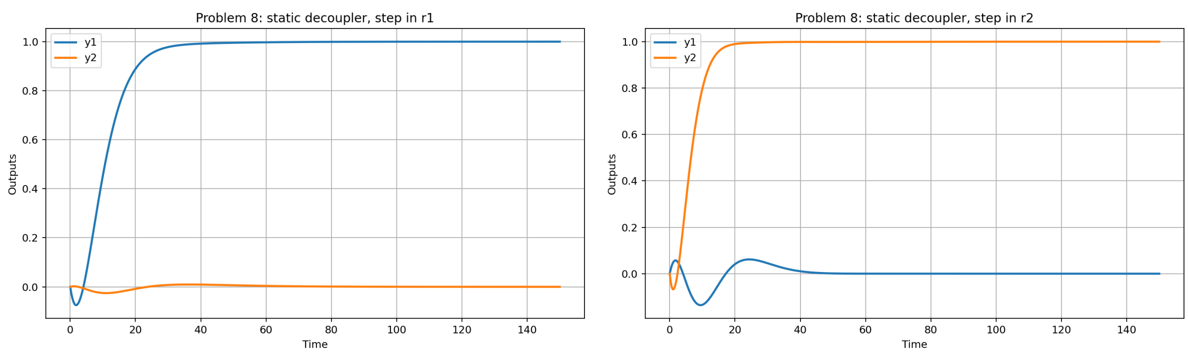
Figure 6: Problem 7 responses with cross terms included.

Problem 8: Static Decouplers

$$D_{12} = -\frac{K_{12}}{K_{11}} = -\frac{2}{5}, \qquad D_{21} = -\frac{K_{21}}{K_{22}} = -\frac{3}{6}$$

Test	Controlled-output IAE	Interaction area
Step in r1	12.4998	0.6320
Step in r2	7.4998	2.1539

Table 11: Problem 8 performance with static decouplers.



(a) Step in r1

(b) Step in r2

Figure 7: Problem 8 responses with static decouplers.

Problem 9: Dynamic Decouplers

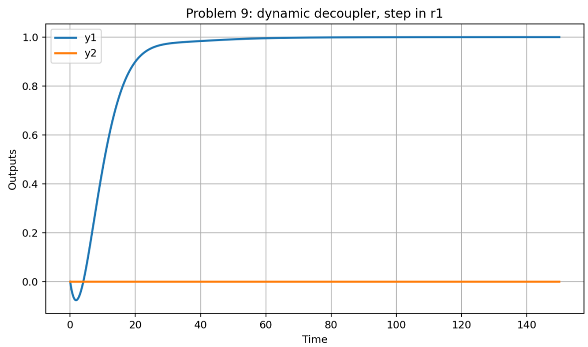
$$D_{12}(s) = -\frac{G_{12}}{G_{11}} = -\frac{2}{5}\frac{4s+1}{8s+1}e^{s}$$
$$D_{21}(s) = -\frac{G_{21}}{G_{22}} = -\frac{3}{6}\frac{10s+1}{12s+1}$$

The term e^{s} in D12 is noncausal because it is equivalent to a negative delay. Therefore, the realizable dynamic decoupler uses only the rational part of D12.

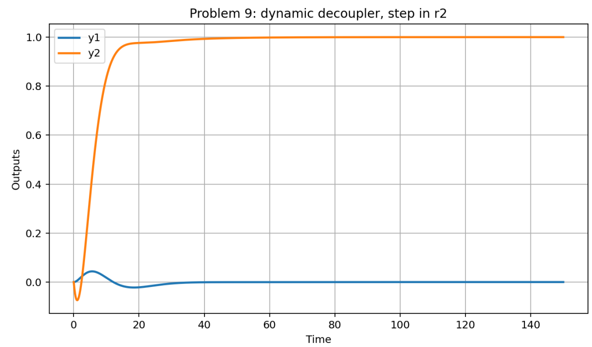
$$D_{12,realizable}(s) = -\frac{2}{5}\frac{4s+1}{8s+1}$$

Test	Controlled-output IAE	Interaction area
Step in r1	12.4998	0.0000
Step in r2	7.4998	0.6073

Table 12: Problem 9 performance with dynamic decouplers.



(a) Step in r1



(b) Step in r2

Figure 8: Problem 9 responses with dynamic decouplers.

Problem 10: Comments on MIMO Results

Case	Step in r1: IAE y1	Step in r1: y2 area	Step in r2: y1 area	Step in r2: IAE y2
No cross terms	10.7171	0.0000	0.0000	6.4302
Cross terms	12.4992	4.5391	4.3600	7.4995
Static decoupler	12.4998	0.6320	2.1539	7.4998
Dynamic decoupler	12.4998	0.0000	0.6073	7.4998

Table 13: Summary of MIMO response metrics for Problems 6--10.

When the cross terms are added, a change in one loop causes movement in the other output, showing loop interaction. Static decouplers reduce the steady-state interaction because they cancel the cross-gains at zero frequency. However, static decoupling cannot remove all transient interaction because the cross terms have different delays and time constants. Dynamic decouplers account for those dynamics and should reduce interaction further, but any noncausal negative-delay terms must be removed or approximated with a realizable transfer function.